
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived causal
structure

Anonymous Author(s)

Affiliation

Address

email

Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ -VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the 20.1M-parameter policy reaches $93.7\% \pm 0.8\%$ with a convolutional encoder under the published three-seed protocol. Averaging three fixed checkpoints reaches 96.2% at three times the inference cost. We derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7} . All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. At the policy’s visual bond, retaining 96 of 384 causally ranked dimensions succeeds in 50/60 matched trials across three checkpoints, compared with 52/60 for the full representation. Three equal-rank random controls reach only 1/60, 3/60, and 3/60. The selected subspace retains 49 of the 52 full-policy successes. On disjoint cache samples, it has $33.5\text{--}110.6\times$ lower reconstruction error than the median random projector across action groups and checkpoints. The evidence establishes competitive specialist capability, exact layerwise access, and a compact causal policy bottleneck. It does not establish broad robotic generalization.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can

rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?

We test two concrete requirements on the same trained policies: competitive closed-loop capability and reproducible structure that has a causal behavioral consequence.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 96-dimensional visual subspace retains 49 of 52 full-policy successes across three checkpoints. Three equal-rank random controls reach only 1/60, 3/60, and 3/60.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes weight-derived structure direct, stable, and behaviorally testable.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

77 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

78 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value,
79 and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike
80 softmax attention, every multiplicative interaction can be expanded into coefficients determined by
81 the trained weights.

82 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
83 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

84 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
85 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
86 operations update this pair. The model therefore keeps input-specific rescaling while remaining
87 exactly representable as a rational tensor network.

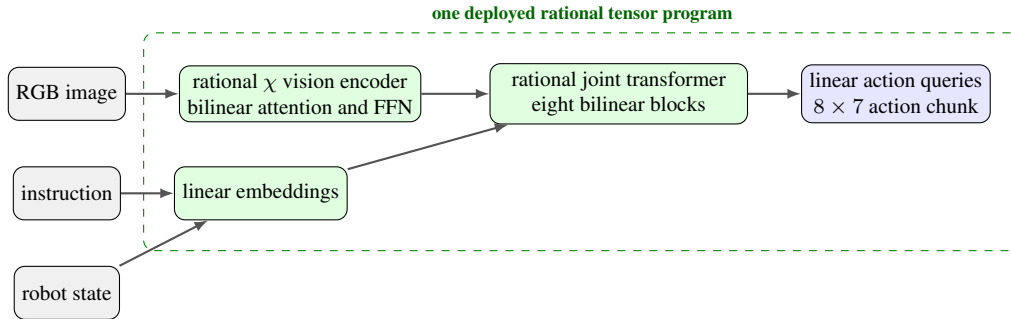


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

88 2.2 Architecture and training

89 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
90 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
91 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
92 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
93 continuous action dimensions. The complete model has 20,137,352 learned parameters.

94 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
95 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
96 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
97 that verify every requested state was loaded.

98 2.3 Mechanical certificate on the trained checkpoint

99 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
100 point at runtime and reconstruct representative operations from saved weights.

Mechanical decomposability audit of the trained 189/200 checkpoint

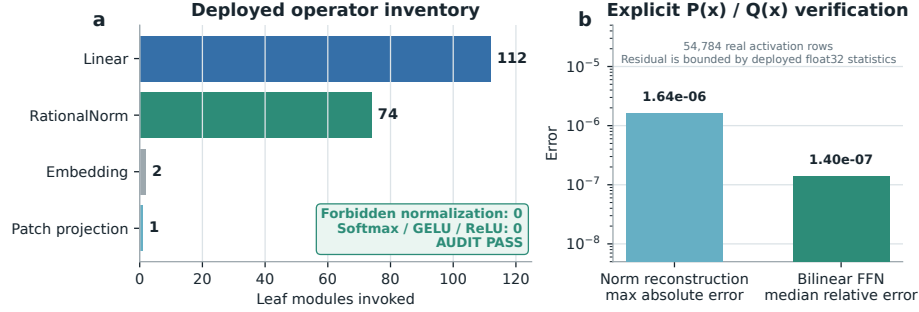


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

101 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 102 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 103 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 104 not materialize one compact polynomial for the full eight-layer transformer.

105 3 Protocol-aligned closed-loop capability

106 3.1 Evaluation protocol

107 We evaluate all ten LIBERO-Object tasks with:

- 108 • 50 canonical trials per task
- 109 • a 280-step cap
- 110 • three independently trained seeds per architecture
- 111 • 500 rollouts per seed and 1,500 rollouts per architecture

112 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 113 rerun in our codebase, so differences may still include implementation and training choices.

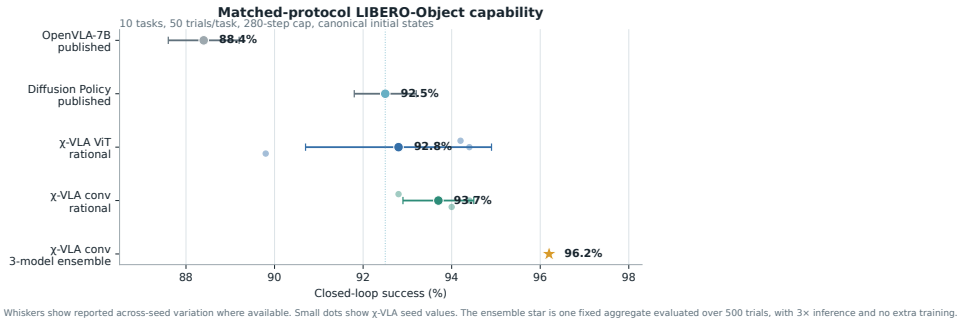


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

114 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

115 The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high
 116 specialist capability is reproduced across three independently trained checkpoints.

117 Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3
 118 points above the mean of its members and 1.2 points above the strongest member. The gain is largest
 119 on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is
 120 not one 20.1M-parameter policy.

121 3.3 What this result does and does not establish

122 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
 123 existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad
 124 robotic generalization or state of the art.

125 4 Exact weight-derived structure through attention

126 4.1 Exact layerwise construction

127 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 128 a high-order coefficient table. Each entry says how strongly one combination of query, key, and
 129 value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe
 130 are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and
 131 an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to
 132 1.24×10^{-14} .

133 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 134 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 135 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 136 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 137 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 138 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 139 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
 140 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

141 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 142 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays
 143 the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most
 144 coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force
 145 materialization to 2.13×10^{-14} .

146 4.2 Trained policy result

147 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
 148 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
 149 sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace.
 150 Real cached activations are used only afterward to measure how faithfully that subspace preserves
 151 the layer’s computation.

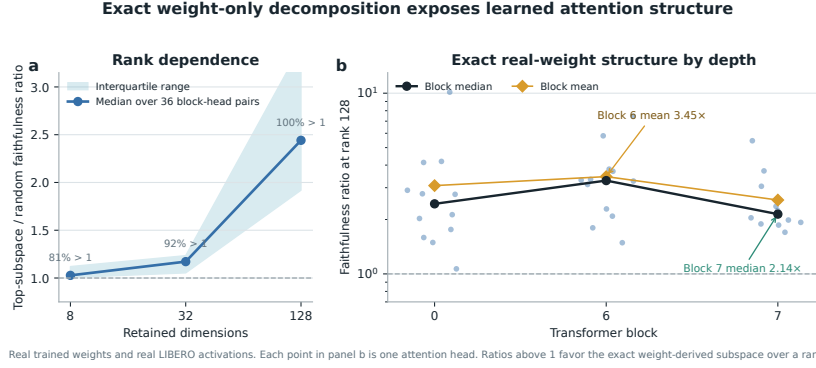


Figure 4: Exact weight-derived attention structure. Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure

We next test whether the identified structure affects closed-loop behavior.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining a subspace means zeroing every direction outside it, then running the actual policy. We compare the selected directions with random equal-width controls in offline reconstruction and closed-loop intervention. This ranking uses cached training demonstrations and downstream action gradients. Offline reconstruction uses a disjoint set of cache samples that did not estimate the ranking. Both splits still come from the training cache, so this is a disjoint-sample in-cache diagnostic. The closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result.

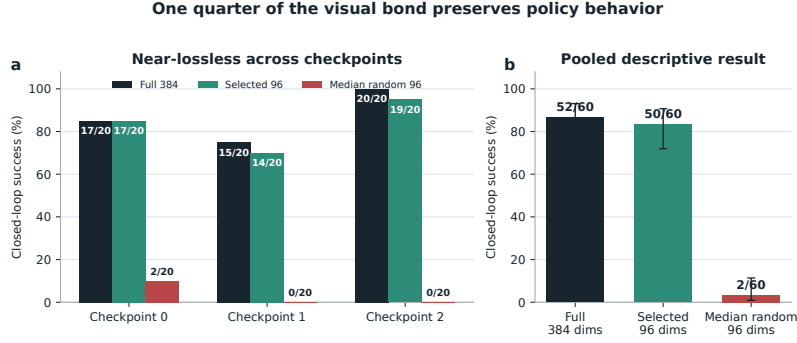


Figure 5: **One quarter of the visual bond preserves policy behavior.** Each checkpoint uses four tasks and five paired canonical trials per task. Rank 96 retains 49 of 52 full-policy successes. Three equal-rank random controls reach 1/60, 3/60, and 3/60. The pooled interval is descriptive because trials within a checkpoint share learned weights.

171 The action-balanced rank screen identifies a 96-dimensional subspace, one quarter of the visual
 172 bond. Full representations succeed in 17/20, 15/20, and 20/20 trials across checkpoints. Selected
 173 subspaces reach 17/20, 14/20, and 19/20, and retain 49 of the 52 exact full-policy successes. Their
 174 conditional retention is 94.1%, 93.3%, and 95.0%. Three random controls per checkpoint total only
 175 1/60, 3/60, and 3/60. Every checkpoint and random-control comparison has the same selected-
 176 over-random direction, with exact paired $p \leq 2.4 \times 10^{-4}$. Offline fidelity uses 1,024 discovery
 177 samples and 1,024 disjoint evaluation samples per checkpoint. Against the median random projector,
 178 the selected basis has 53.0–110.6 $\times$ lower translation error, 37.5–62.1 $\times$ lower rotation error, and
 179 33.5–103.0 $\times$ lower gripper error. This repairs sample reuse, but it remains an in-cache fidelity test
 180 rather than held-out task generalization. This data-driven full-policy ranking complements the exact
 181 layerwise analysis in Section 4.

182 6 Related work

183 **VLA capability.** OpenVLA established an open pretrained VLA and standardized LIBERO com-
 184 parisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can
 185 substantially improve adaptation and control efficiency [2]. These systems use large pretrained
 186 backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

187 **Tensor and polynomial networks.** Polynomial networks and tensor networks use explicit mul-
 188 tilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight
 189 eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for com-
 190 positional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an
 191 exact reduced construction for individual softmax-free attention layers.

192 **Causal mechanisms and policy analysis.** Bilinear IMPALA policies have combined competitive
 193 ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5].
 194 This is the closest architectural precedent for weight-based policy analysis. Recent work steers
 195 pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify
 196 and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions
 197 recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These
 198 results make a universal separation from conventional policies untenable. Our intended distinction
 199 is empirical: exact weight-derived discovery and a replicated causal bottleneck in a competitive
 200 continuous vision-language policy.

201 **VLA robustness.** Recent evaluations show that standard LIBERO success can coexist with weak
 202 robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist
 203 capability setting rather than evidence of broad generalization.

7 Limitations and conclusion

χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers, and a compact causal visual subspace retains 49 of 52 full-policy successes across three checkpoints while equal-rank random controls rarely succeed.

The scope is deliberately specific. LIBERO-Object is one specialist suite, so these results do not establish broad robotic generalization. Exact reconstruction is certified layer by layer, while the full-policy visual ranking also uses cached activations and action gradients. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

Within that scope, polynomial and rational restrictions are compatible with competitive closed-loop control and provide a precise weight-derived analysis surface. The same trained policies support the capability, exact reconstruction, and causal-bottleneck results, linking the algebraic model artifact to behavior rather than treating decomposability as an architectural promise alone.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Doods, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Doods, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Erisken, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

A.2 Per-task ensemble behavior

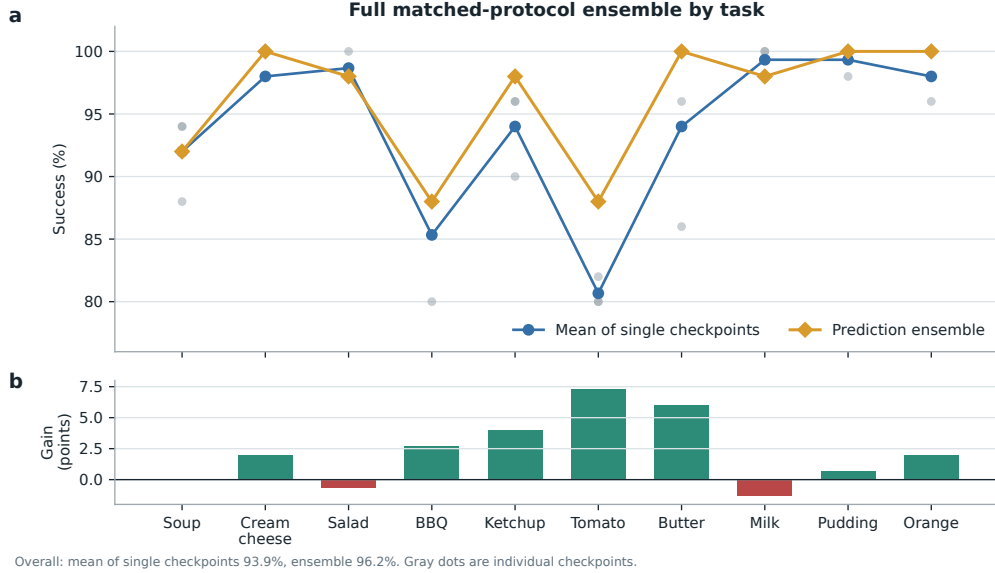


Figure 6: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.3 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.4 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.